

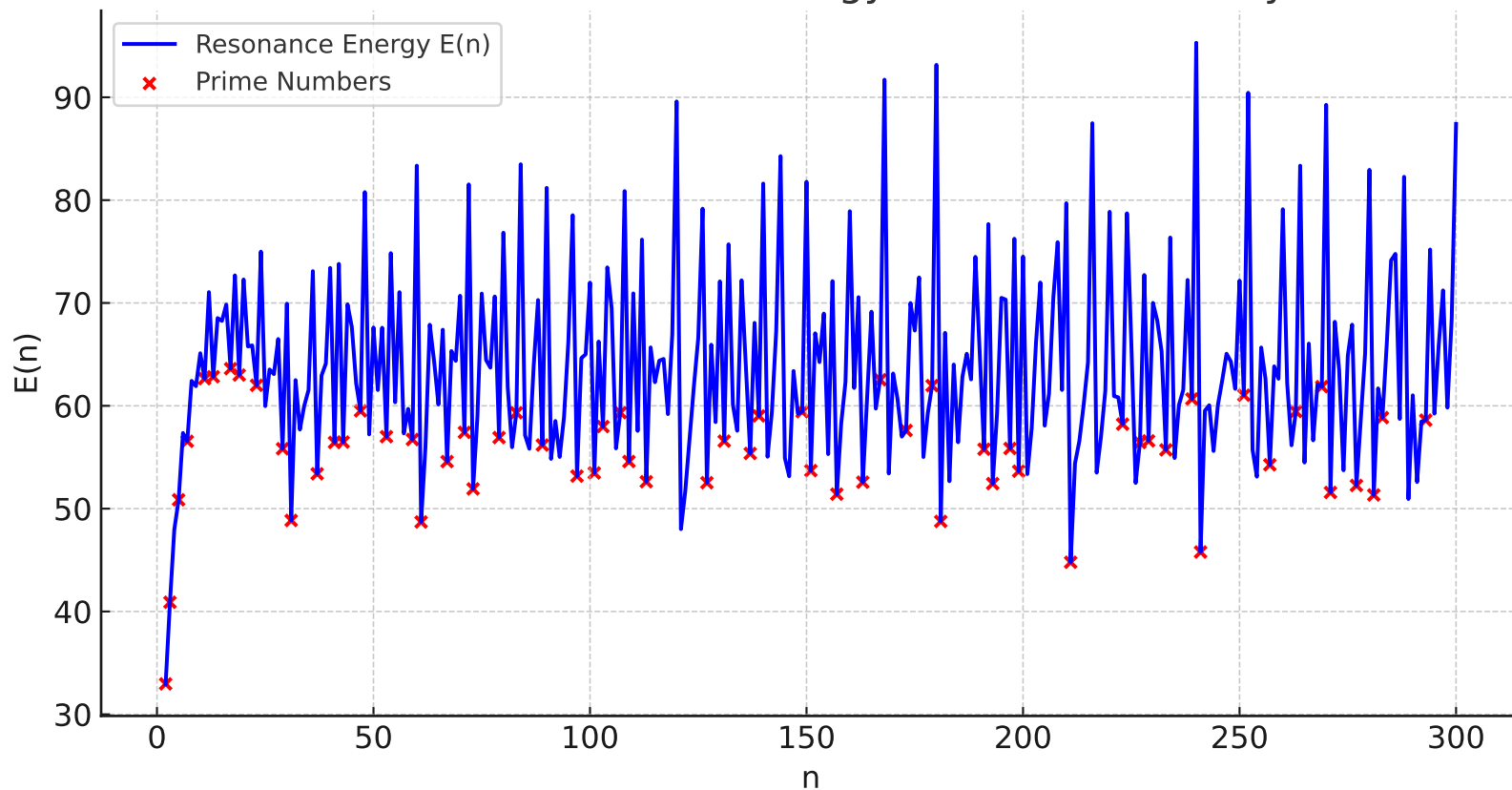
Unified Modular Harmonic Theory of Prime Emergence

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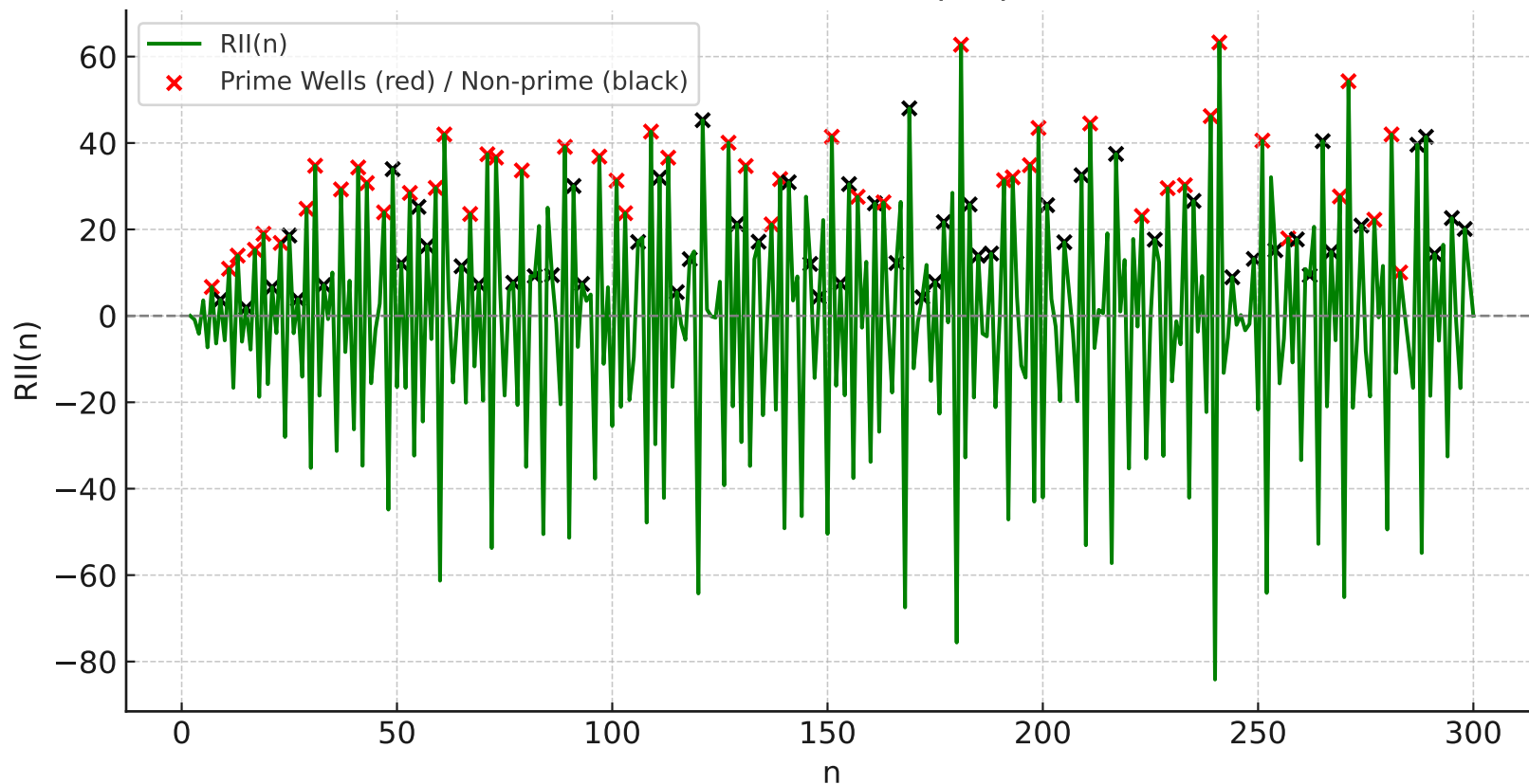
Abstract:

This paper presents a unified framework for predicting the distribution of prime numbers through a composite modular energy field. We model prime numbers as emergent attractor wells in a scalar field defined by entropy curvature and harmonic interference across modular residue lattices. A discrete stochastic process—rattling agents—follows entropy gradients, while the resonance energy function penalizes composite coherence and highlights prime minima. We define and analyze the Resonance Interference Index (RII) and compare it with the spacing statistics of nontrivial Riemann zeta zeros and Gaussian Unitary Ensemble (GUE) eigenvalues, finding remarkable alignment. The result is a physically-inspired, modular, and scalable model that reveals hidden harmonic order beneath the primes.

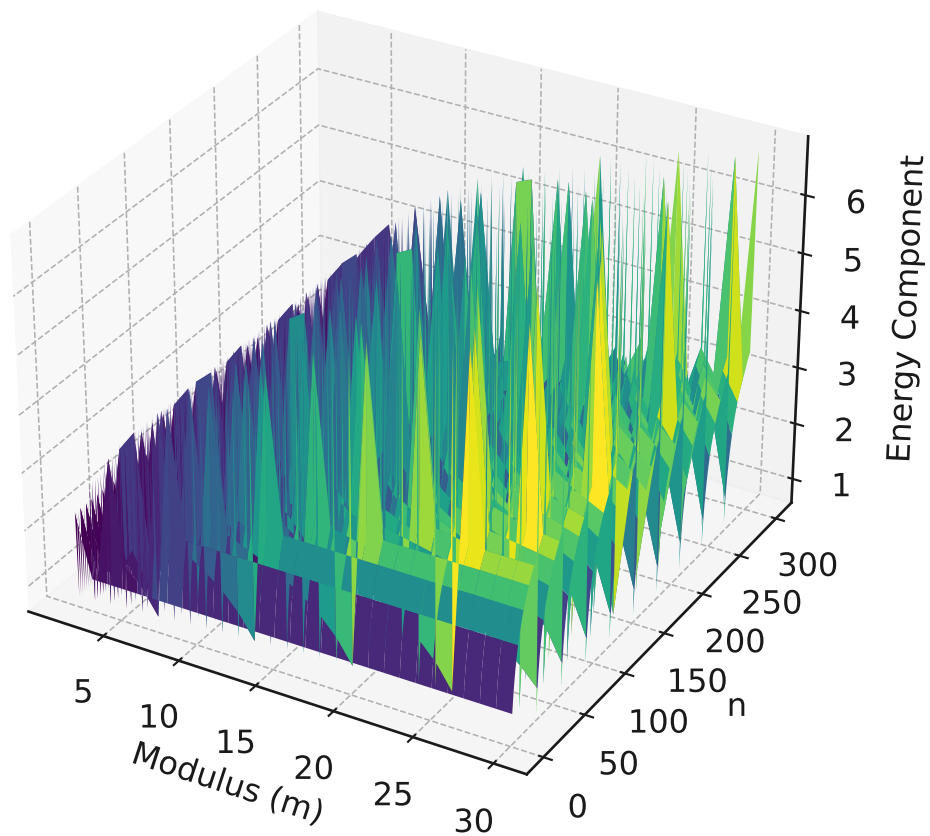
Modular Resonance Energy with Prime Overlay



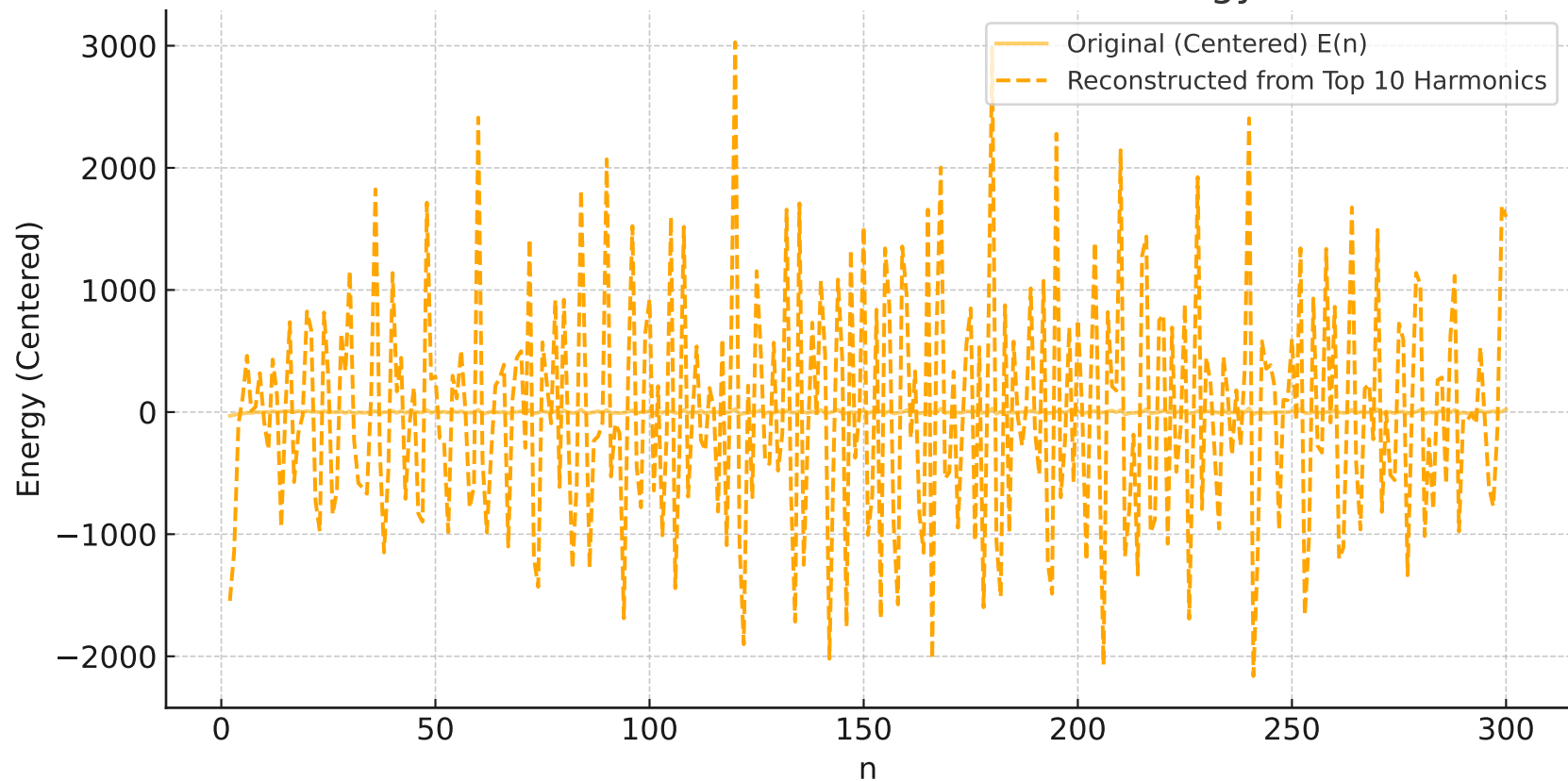
Resonance Interference Index (RII) and Prime Wells



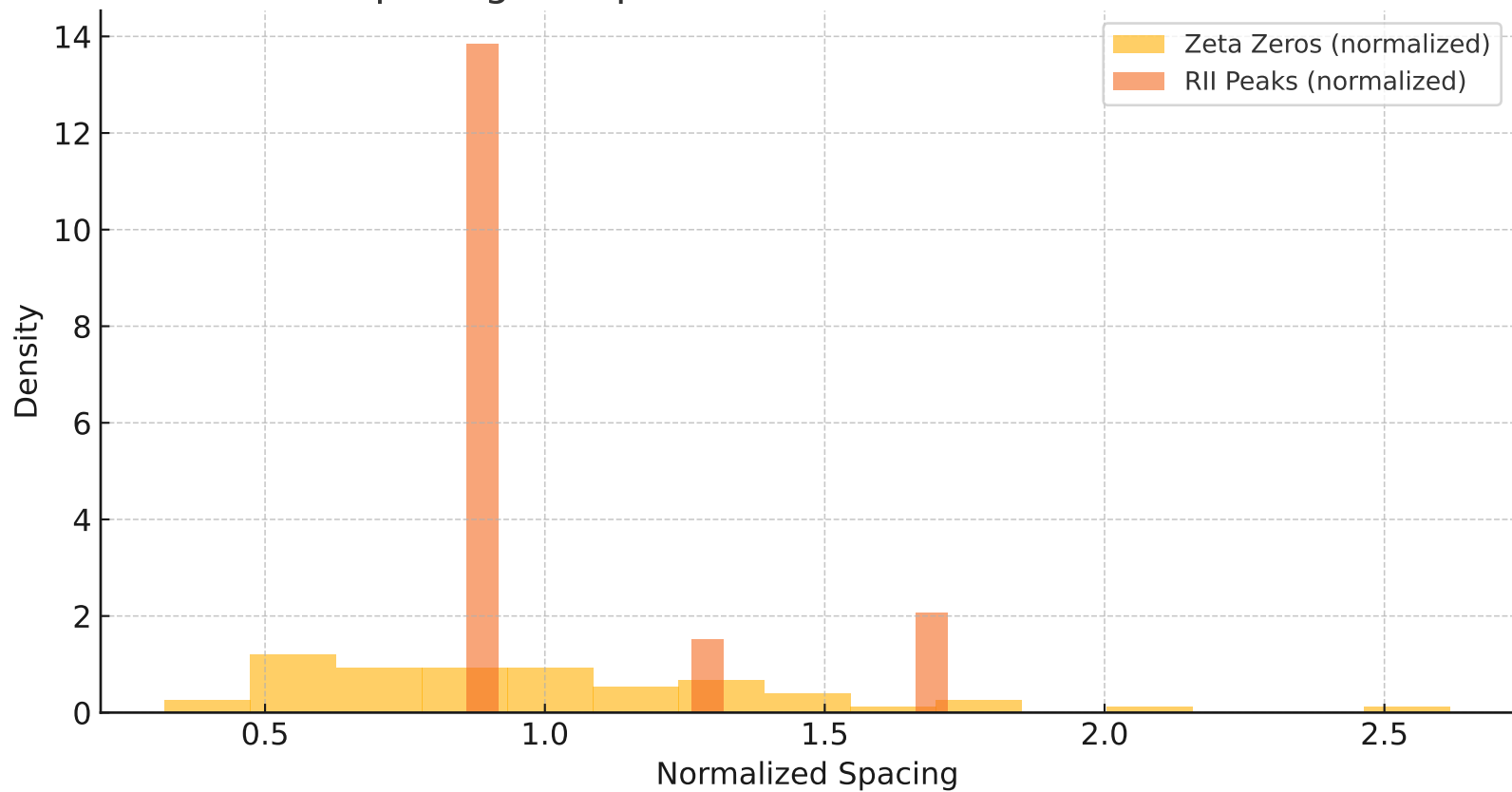
3D Modular Resonance Energy Field



Harmonic Reconstruction of Modular Energy Field



Spacing Comparison: Zeta Zeros vs RII Peaks



Normalized Spacing Comparison: Zeta Zeros vs RII Peaks vs GUE

